Doc Number: 161310G00000-00	Doc Version: 008	Date: 02-Nov-2026

Super Mushshak Engine Acceleration Automatic Test

Prepar3D v5 | Rahwali Airfield (OPRW) | Automatic Testing Mode

Super Mushshak Logger - ver. 1.0

Super Mushshak Engine Automatic Testing Suite

Flight Test File: Super_Mushak_auto_takeoff_data_2026-11-02_12-28-34_legacy_plot.csv

ONLINE, AUTOMATIC

Test Date: 02-Nov-2026 Test Time: 12:28:34

Initial Recorded Conditions

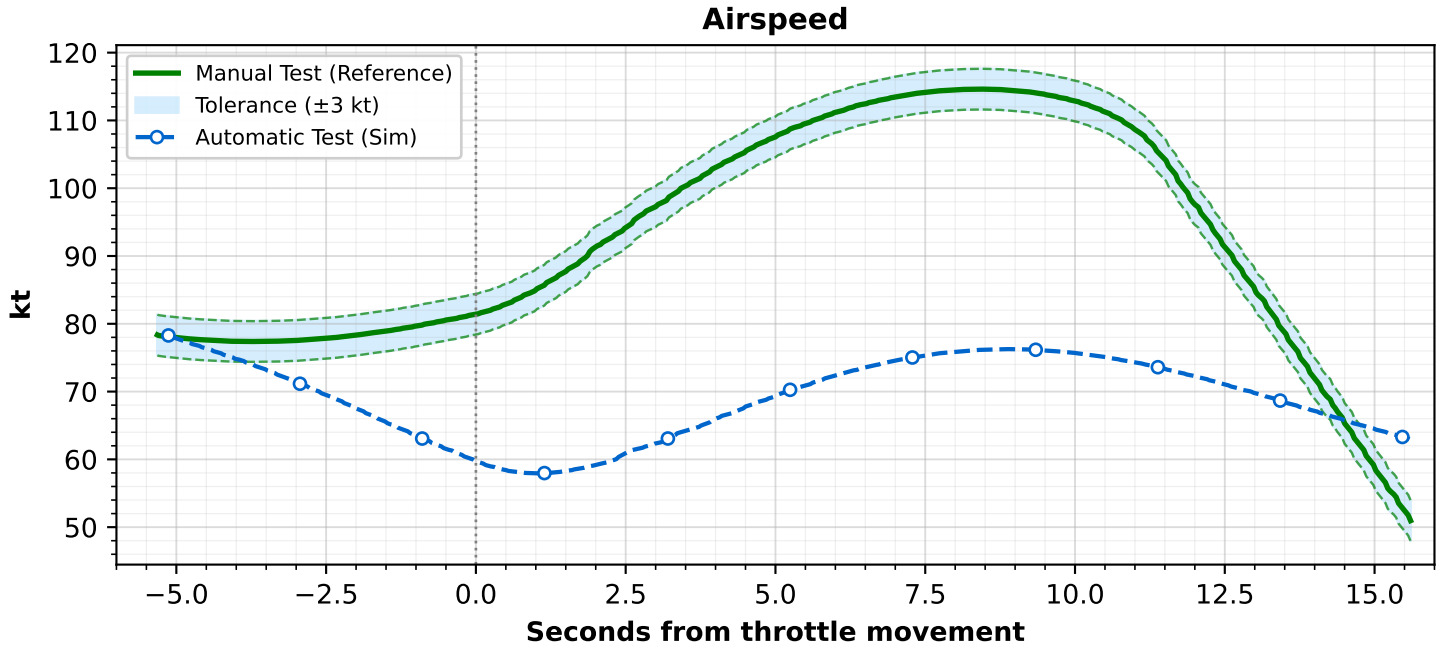
Validation	Sim	
***** Engine Parameters (Piston Engine) *****		
Engine 1		
5.00	5.00	Throttle Position (%)
1505	1494	Engine Speed (RPM)
1505	1494	Propeller Speed (RPM)
5.37	5.38	Manifold Pressure (inHg)
2.44	2.44	Fuel Flow (lb/h)

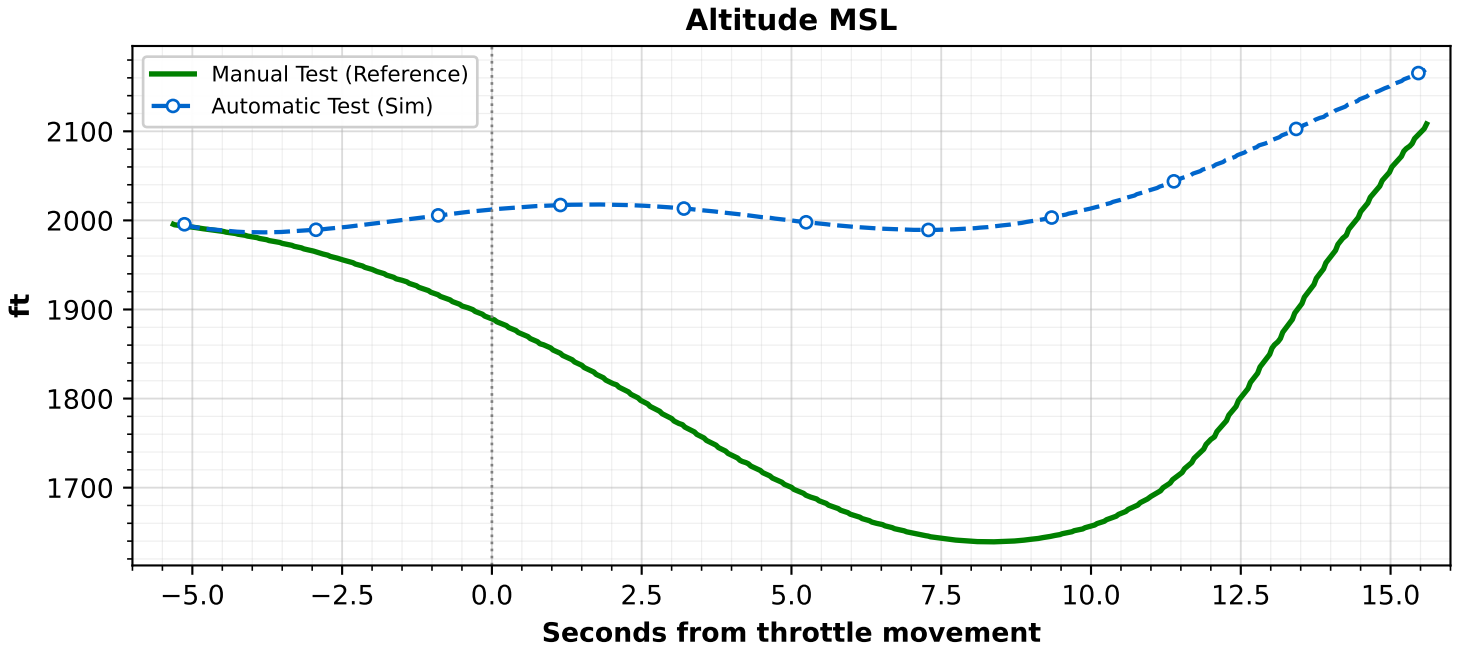
Validation	Sim	
***** Flight Parameters *****		
78.4	78.3	Indicated Airspeed (kt)
1996	1996	Altitude MSL (ft)
1251	1251	Height AGL (ft)
***** Environment Parameters *****		
1013	1013	Baro Press (hPa)
14.3	14.3	OAT (deg C)
0.00	0.00	Wind Dir (mag,deg)
0.00	0.00	Wind Spd (kt)
***** Sim Systems *****		
Off	Off	Primary Motion (-)
Off	Off	Secondary Motion (-)
-	-	Sound Level (-)
50	50	Vibration Level (%)

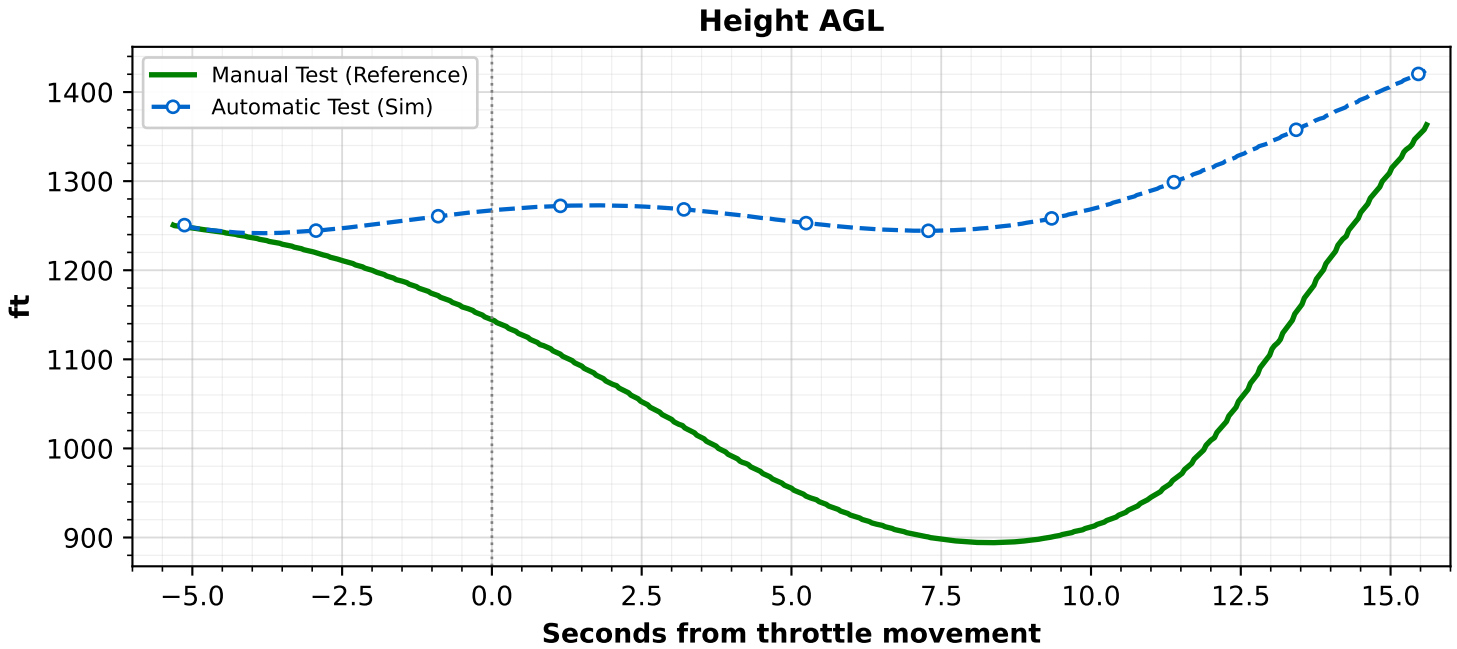
Time zero: measured throttle movement in each run.

Source: Super_Mushak_auto_takeoff_data_2026-11-02_12-28-34_legacy_plot.csv

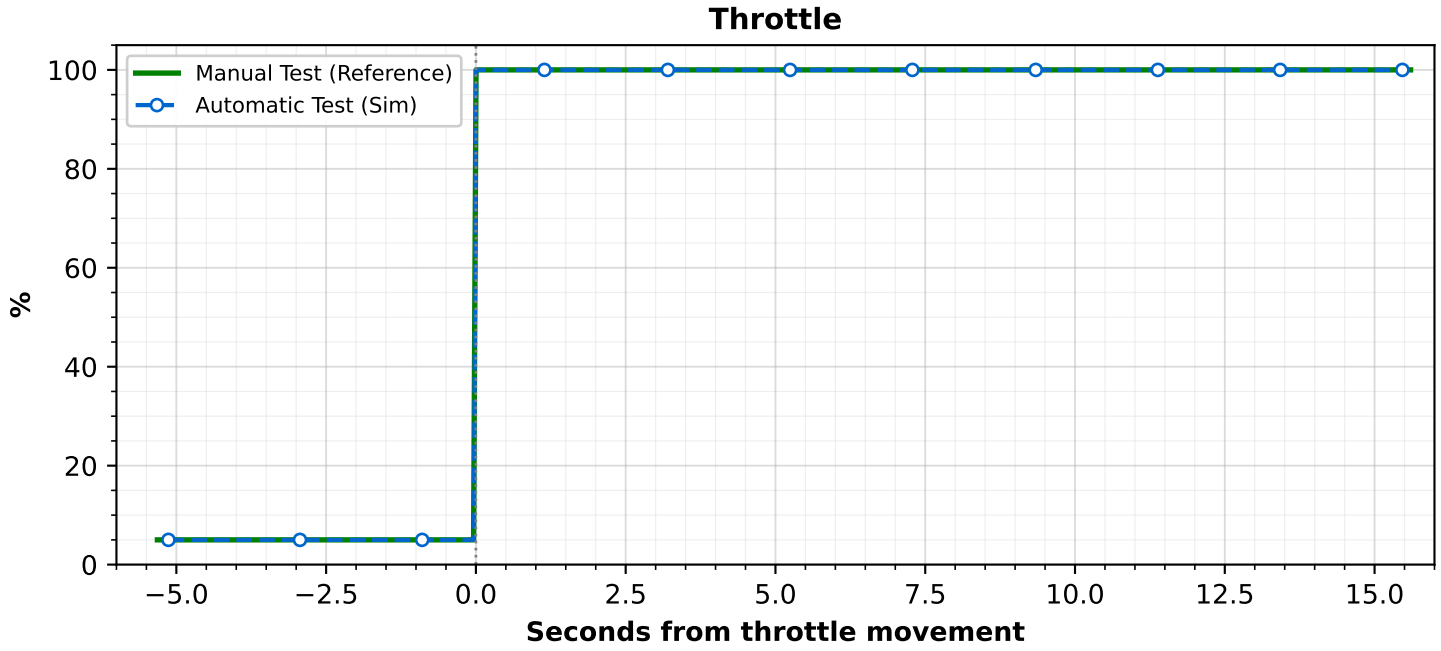
1st Quadrant Combined Group 1 - Airspeed and Altitude

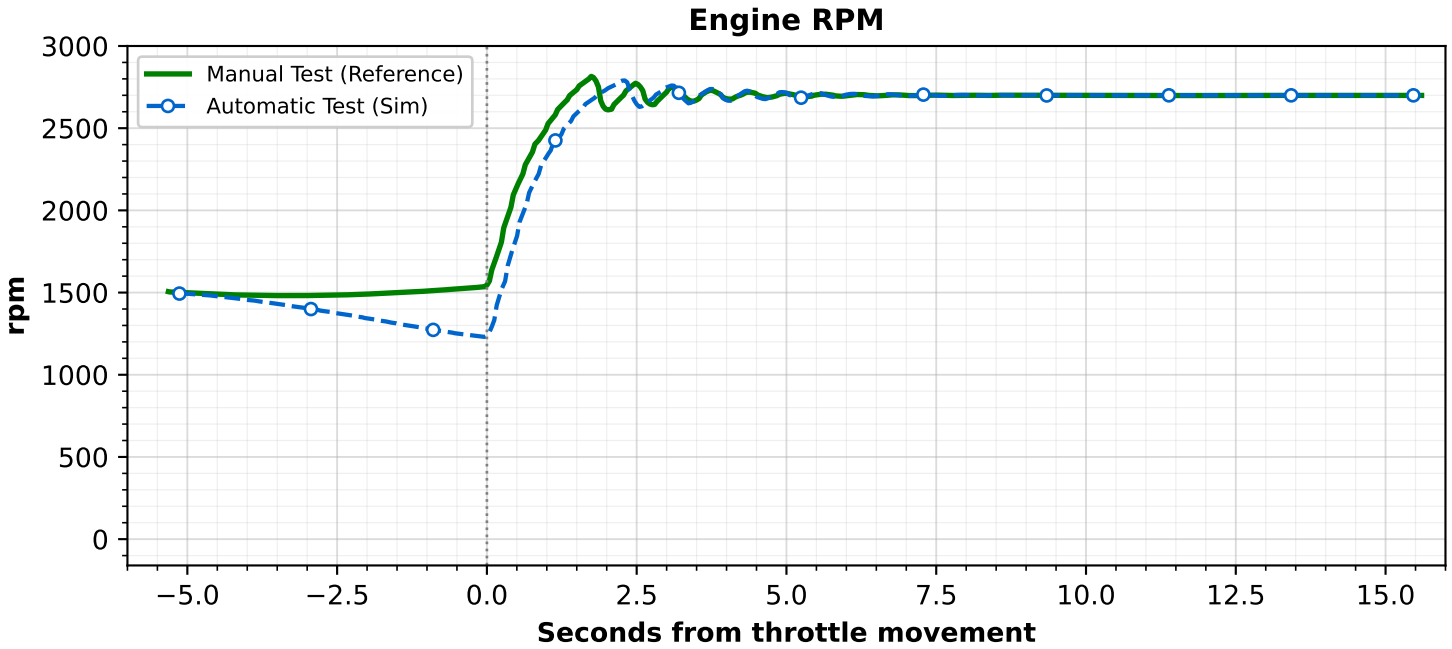


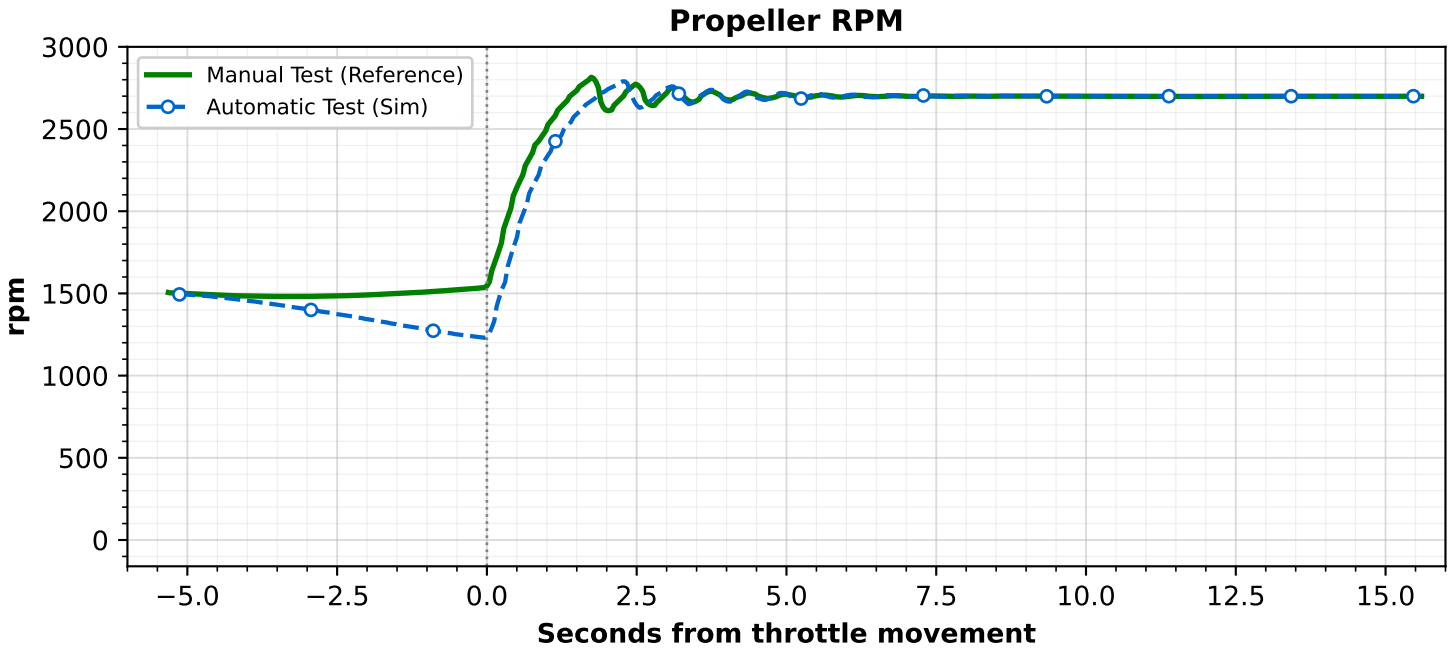




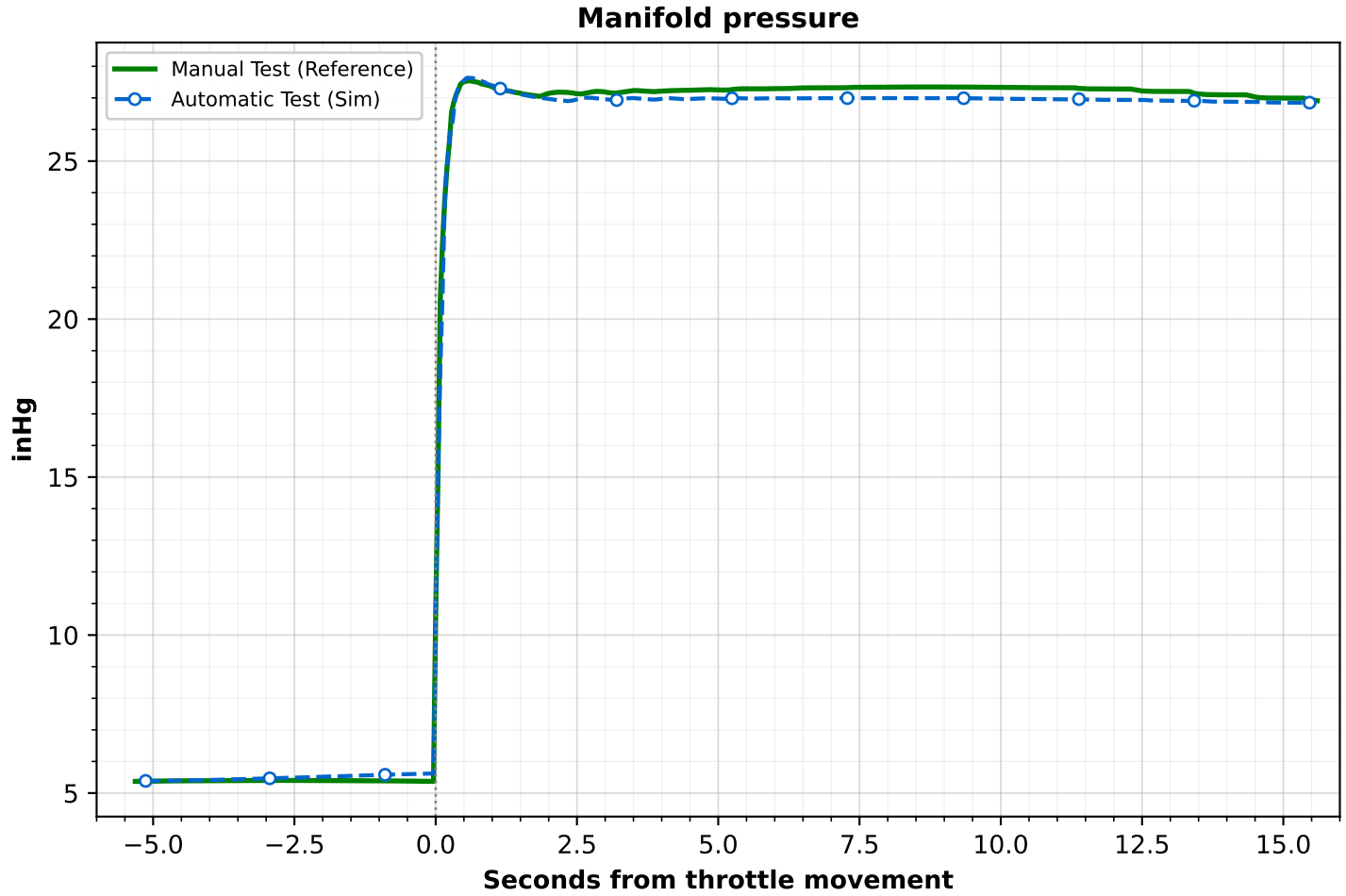
1st Quadrant Combined Group 2 - Throttle and Engine Response

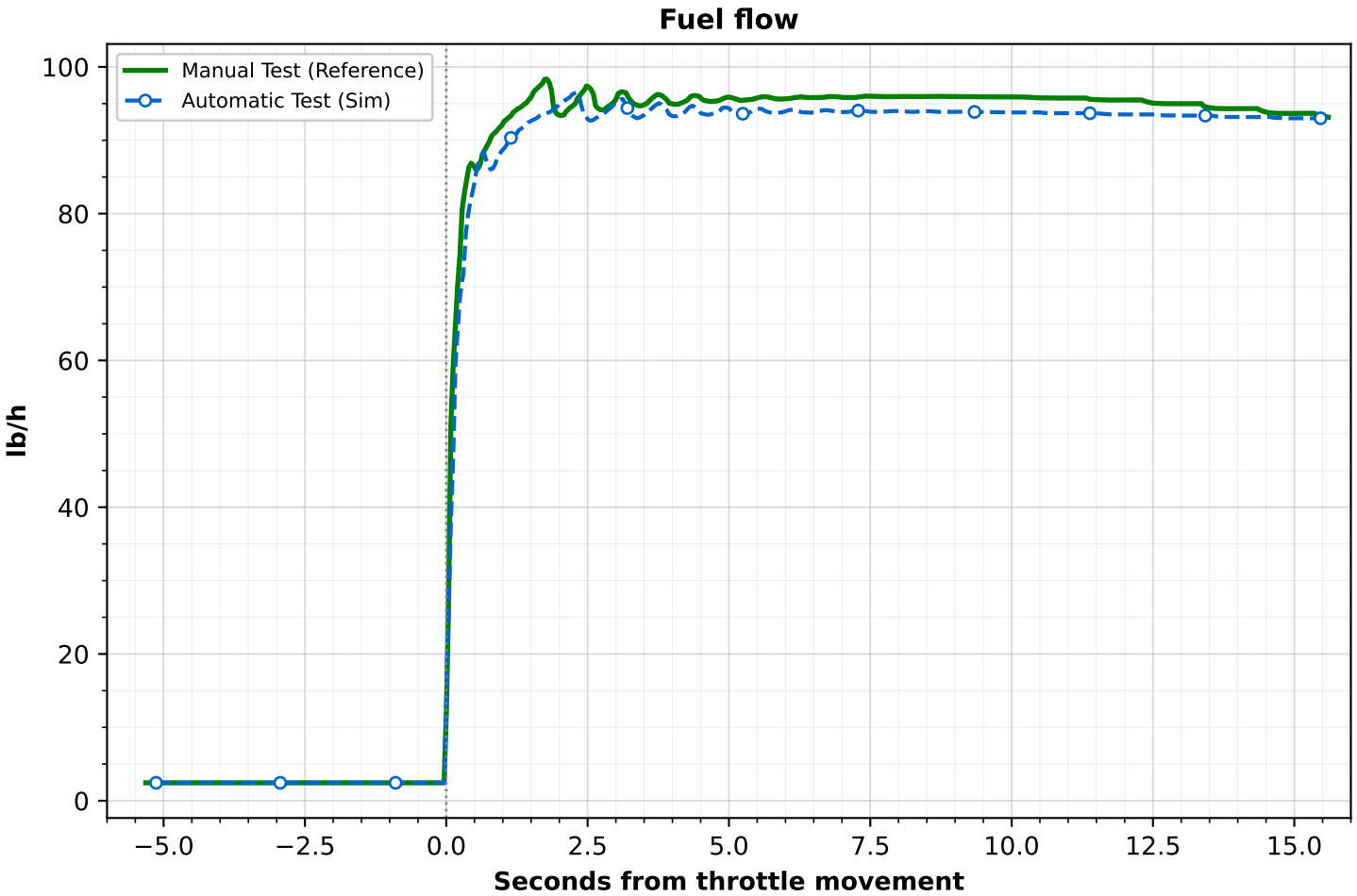






1st Quadrant Combined Group 3 - Engine Power Parameters





Super Mushshak Engine Acceleration - Ti / Tt Response Analysis
Engineering timing check - CS-FSTD(A) Issue 2, 1.f method

Metric	Manual	Automatic	Difference	Compare band	Result
Ti (10% response)	0.015 s	0.014 s	-0.001 s	+/-0.25 s	**Within**
Tt (90% response)	0.245 s	0.231 s	-0.014 s	+/-0.02 s	**Within**

Timing/repeatability: WITHIN_TOLERANCE

Channel: Manifold pressure (inHg)

Comparison bands: Ti = max(10% of Manual Ti, 0.25 s); Tt = 10% of Manual Tt (engineering comparison only).

